Proof Strategies-I

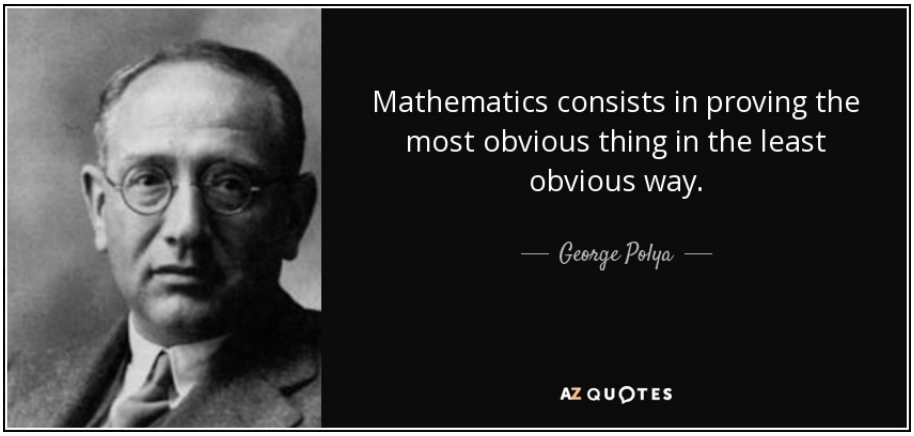
In this lecture we will learn about:

- Why Proofs are needed?
- Proofs from Childhood.
- Rules of inference.
- Axiomatic Systems and Theory of proofs.

Lecture Credo

"I see, I forget.
I hear, I remember.
I do, I understand."

— Confucius



Readings:

Discrete Mathematics and Its Applications Seventh Edition(Kenneth H. Rosen): Page 80-107

Student Name: _____

Student ID: _____

Semester: _____

***Welcome*,** to new topic of discrete mathematics.
Remember our main anchor of the course is to treat mathematics as **GAME.**

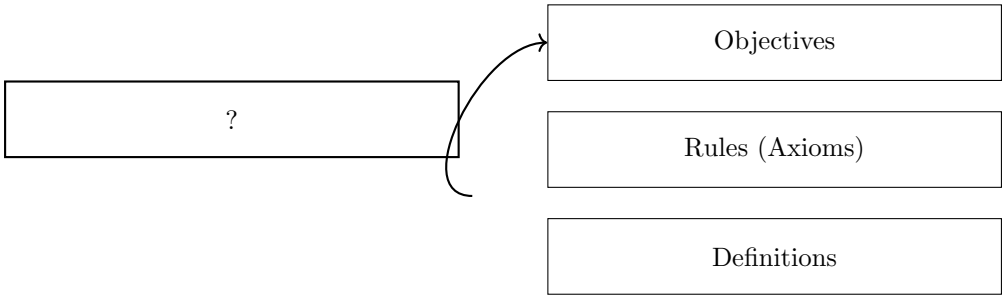


Figure (*): GAME OF MATHEMATICS

Self Reflection

a) Does knowing all the definitions of the GAME make you a good player?

☐ Yes ☐ No

b) Does knowing all the rules of the GAME make you a good player?

☐ Yes ☐ No

If you have doubt, think of players like ***Sachin Tendulkar, Virat Kohli, Cristiano Ronaldo*** in their games.

c) What makes somebody a good(or even a great) player?

Ex. *Write your answer below*

The missing piece in the above ***Figure (*)*** is

STRATEGY

Without strategy one cannot understand, play and excel at the game.

In our ***Game of Mathematics***,

STRATEGY is PROOFS

The roots of this understanding stem from ***Logic***.

Self Reflection

a) Have you started reading the book, **"Discrete Mathematics and Its Applications by Kenneth H. Rosen"**?

☐ Yes ☐ No

b) How many pages have you read so far?

Ex. *Write your answer below*

c) How much time it took you to read the above mentioned pages?

Ex. *Write your answer below*

Let's try reading, learning and understanding right now right here!

1.6 Rules of Inference

Introduction

Later in this chapter we will study proofs. Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an **argument**, we mean a sequence of statements that end with a conclusion. By **valid**, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or **premises**, of the argument. That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false. To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements.

Before we study mathematical proofs, we will look at arguments that involve only compound propositions. We will define what it means for an argument involving compound propositions to be valid. Then we will introduce a collection of rules of inference in propositional logic. These rules of inference are among the most important ingredients in producing valid arguments. After we illustrate how rules of inference are used to produce valid arguments, we will describe some common forms of incorrect reasoning, called **fallacies**, which lead to invalid arguments.

After studying rules of inference in propositional logic, we will introduce rules of inference for quantified statements. We will describe how these rules of inference can be used to produce valid arguments. These rules of inference for statements involving existential and universal quantifiers play an important role in proofs in computer science and mathematics, although they are often used without being explicitly mentioned.

Finally, we will show how rules of inference for propositions and for quantified statements can be combined. These combinations of rule of inference are often used together in complicated arguments.

Valid Arguments in Propositional Logic

Consider the following argument involving propositions (which, by definition, is a sequence of propositions):

"If you have a current password, then you can log onto the network."

"You have a current password."

Therefore,

"You can log onto the network."

We would like to determine whether this is a valid argument. That is, we would like to determine whether the conclusion "You can log onto the network" must be true when the premises "If you have a current password, then you can log onto the network" and "You have a current password" are both true.

Before we discuss the validity of this particular argument, we will look at its form. Use p to represent "You have a current password" and q to represent "You can log onto the network." Then, the argument has the form

$$\begin{array}{l} p \to q \\ p \\ \hline \therefore q \end{array}$$

where $\therefore$ is the symbol that denotes "therefore."

We know that when p and q are propositional variables, the statement $((p \to q) \wedge p) \to q$ is a tautology (see Exercise 10(c) in Section 1.3). In particular, when both $p \to q$ and p are true, we know that q must also be true. We say this form of argument is **valid** because whenever all its premises (all statements in the argument other than the final one, the conclusion) are true, the conclusion must also be true. Now suppose that both "If you have a current password, then you can log onto the network" and "You have a current password" are true statements. When we replace p by "You have a current password" and q by "You can log onto the network," it necessarily follows that the conclusion "You can log onto the network" is true. This argument is **valid** because its form is valid. Note that whenever we replace p and q by propositions where $p \to q$ and p are both true, then q must also be true.

What happens when we replace p and q in this argument form by propositions where not both p and $p \to q$ are true? For example, suppose that p represents "You have access to the network" and q represents "You can change your grade" and that p is true, but $p \to q$ is false. The argument we obtain by substituting these values of p and q into the argument form is

"If you have access to the network, then you can change your grade."
"You have access to the network."

$\therefore$ "You can change your grade."

The argument we obtained is a valid argument, but because one of the premises, namely the first premise, is false, we cannot conclude that the conclusion is true. (Most likely, this conclusion is false.)

In our discussion, to analyze an argument, we replaced propositions by propositional variables. This changed an argument to an **argument form**. We saw that the validity of an argument follows from the validity of the form of the argument. We summarize the terminology used to discuss the validity of arguments with our definition of the key notions.

DEFINITION 1 An *argument* in propositional logic is a sequence of propositions. All but the final proposition in the argument are called *premises* and the final proposition is called the *conclusion*. An argument is *valid* if the truth of all its premises implies that the conclusion is true.

An *argument form* in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is *valid* no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion q is valid, when $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \to q$ is a tautology.

The key to showing that an argument in propositional logic is valid is to show that its argument form is valid. Consequently, we would like techniques to show that argument forms are valid. We will now develop methods for accomplishing this task.

Rules of Inference for Propositional Logic

We can always use a truth table to show that an argument form is valid. We do this by showing that whenever the premises are true, the conclusion must also be true. However, this can be a tedious approach. For example, when an argument form involves 10 different propositional variables, to use a truth table to show this argument form is valid requires $2^{10} = 1024$ different rows. Fortunately, we do not have to resort to truth tables. Instead, we can first establish the validity of some relatively simple argument forms, called **rules of inference**. These rules of inference can be used as building blocks to construct more complicated valid argument forms. We will now introduce the most important rules of inference in propositional logic.

The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the rule of inference called **modus ponens**, or the **law of detachment**. (Modus ponens is Latin for *mode that affirms*.) This tautology leads to the following valid argument form, which we have already seen in our initial discussion about arguments (where, as before, the symbol $\therefore$ denotes "therefore"):

$$\begin{array}{l} \quad p \\ \quad \underline{p \rightarrow q} \\ \therefore q \end{array}$$

Using this notation, the hypotheses are written in a column, followed by a horizontal bar, followed by a line that begins with the therefore symbol and ends with the conclusion. In particular, modus ponens tells us that if a conditional statement and the hypothesis of this conditional statement are both true, then the conclusion must also be true. Example 1 illustrates the use of modus ponens.

EXAMPLE 1 Suppose that the conditional statement "If it snows today, then we will go skiing" and its hypothesis, "It is snowing today," are true. Then, by modus ponens, it follows that the conclusion of the conditional statement, "We will go skiing," is true. ◀

As we mentioned earlier, a valid argument can lead to an incorrect conclusion if one or more of its premises is false. We illustrate this again in Example 2.

EXAMPLE 2 Determine whether the argument given here is valid and determine whether its conclusion must be true because of the validity of the argument.

"If $\sqrt{2} > \frac{3}{2}$, then $\left(\sqrt{2}\right)^2 > \left(\frac{3}{2}\right)^2$. We know that $\sqrt{2} > \frac{3}{2}$. Consequently, $\left(\sqrt{2}\right)^2 = 2 > \left(\frac{3}{2}\right)^2 = \frac{9}{4}$."

Solution: Let p be the proposition "$\sqrt{2} > \frac{3}{2}$" and q the proposition "$2 > (\frac{3}{2})^2$." The premises of the argument are $p \rightarrow q$ and p, and q is its conclusion. This argument is valid because it is constructed by using modus ponens, a valid argument form. However, one of its premises, $\sqrt{2} > \frac{3}{2}$, is false. Consequently, we cannot conclude that the conclusion is true. Furthermore, note that the conclusion of this argument is false, because $2 < \frac{9}{4}$. ◀

There are many useful rules of inference for propositional logic. Perhaps the most widely used of these are listed in Table 1. Exercises 9, 10, 15, and 30 in Section 1.3 ask for the verifications that these rules of inference are valid argument forms. We now give examples of arguments that use these rules of inference. In each argument, we first use propositional variables to express the propositions in the argument. We then show that the resulting argument form is a rule of inference from Table 1.

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

EXAMPLE 3 State which rule of inference is the basis of the following argument: "It is below freezing now. Therefore, it is either below freezing or raining now."

Solution: Let p be the proposition "It is below freezing now" and q the proposition "It is raining now." Then this argument is of the form

$$\begin{array}{l} \quad p \\ \hline \therefore p \vee q \end{array}$$

This is an argument that uses the addition rule. ◀

EXAMPLE 4 State which rule of inference is the basis of the following argument: "It is below freezing and raining now. Therefore, it is below freezing now."

Solution: Let p be the proposition "It is below freezing now," and let q be the proposition "It is raining now." This argument is of the form

$$\begin{array}{l} \quad p \wedge q \\ \hline \therefore p \end{array}$$

This argument uses the simplification rule. ◀

EXAMPLE 5 State which rule of inference is used in the argument:

> If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Solution: Let p be the proposition "It is raining today," let q be the proposition "We will not have a barbecue today," and let r be the proposition "We will have a barbecue tomorrow." Then this argument is of the form

$$\begin{array}{rl} & p \to q \\ & q \to r \\ \hline \therefore & p \to r \end{array}$$

Hence, this argument is a hypothetical syllogism. ◀

Using Rules of Inference to Build Arguments

When there are many premises, several rules of inference are often needed to show that an argument is valid. This is illustrated by Examples 6 and 7, where the steps of arguments are displayed on separate lines, with the reason for each step explicitly stated. These examples also show how arguments in English can be analyzed using rules of inference.

EXAMPLE 6 Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Extra Examples

Solution: Let p be the proposition "It is sunny this afternoon," q the proposition "It is colder than yesterday," r the proposition "We will go swimming," s the proposition "We will take a canoe trip," and t the proposition "We will be home by sunset." Then the premises become $\neg p \wedge q$, $r \to p$, $\neg r \to s$, and $s \to t$. The conclusion is simply t. We need to give a valid argument with premises $\neg p \wedge q$, $r \to p$, $\neg r \to s$, and $s \to t$ and conclusion t.

We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	**Reason**
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \to p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \to s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \to t$	Premise
8. t	Modus ponens using (6) and (7)

Note that we could have used a truth table to show that whenever each of the four hypotheses is true, the conclusion is also true. However, because we are working with five propositional variables, p, q, r, s, and t, such a truth table would have 32 rows. ◀

If you have read it with focus then, try to figure out the following questions.

a) Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

Ex. *Write your answer below*

b) Determine whether the following argument is correct or incorrect and explain why? **"Quincy likes all action movies. Quincy likes the movie Eight Men Out. Therefore, Eight Men Out is an action movie."**

Ex. *Write your answer below*

c) **What is wrong with this argument?** Let $H(x)$ be "x is happy." Given the premise $\exists x\, H(x)$, we conclude that $H(\text{Harry})$. Therefore, Harry is happy.

Ex. *Write your answer below*

Let's come back to learning ***STRATEGY of the GAME OF MATHEMATICS, THE ART OF WRITING PROOFS.***

1 The First Proof

- **Demonstrative math = Proof-based math.**
  Before: *What is true.* After: *Why it is true.*

- **Thales of Miletus** ($\sim$ 624–545 BC, from Miletus) was a Greek mathematician.
- Thales is the first person to whom a mathematical proof is attributed.
- He's also called the Father of Science because he used evidence and theories, not mythology, to study and explain the world.

Theorem: The diameter of any circle divides that circle into two equal halves.

Ex. *Write your proof*

Activity

Pull the last Activity Page and follow the instructions.

2 Let's recall our memories to prove mathematical statements!

Theorem. There is no rational number x which satisfies $x^2 = 2$.

The above statement of the theorem can be written in several equivalent ways. For example:

- **Theorem.** $\sqrt{2}$ is irrational.
- **Theorem.** $\forall x \in \mathbb{Q},\ x^2 = 2$ has no solution.
- **Theorem.** $\nexists x \in \mathbb{Q}$ such that $x^2 = 2$.

Ex. *Write your proof*

3 Axiomatic Systems

Definition

An *axiomatic system* is a finite sequence of propositions $a_1, a_2, \ldots, a_N$, which are called the *axioms* of the system.

4 Definition of the Proof

A *proof* of a proposition p within an axiomatic system $a_1, a_2, \ldots, a_N$ is a finite sequence of propositions $q_1, q_2, \ldots, q_M$ such that $q_M = p$ and for any $1 \leq j \leq M$ **one** of the following is satisfied:

(A) q_j is a proposition from the list of axioms;

(T) q_j is a tautology;

(M) $\exists\, 1 \leq m, n < j : (q_m \wedge q_n \Rightarrow q_j)$ is true.

(a) What is the mean by tautology?

Ex. *Write your answer below*

Remarks

Remark 1. If p can be proven within an axiomatic system $a_1, a_2, \ldots, a_N$, we write:

$$a_1, a_2, \ldots, a_N \vdash p$$

and we read "$a_1, a_2, \ldots, a_N$ **proves** p".

Remark 2. This definition allows us to easily recognize a proof. A computer could easily check whether or not the conditions (A), (T), and (M) are satisfied by a given sequence of propositions. To actually find a proof of a proposition is a whole different story.

Remark 3. Obviously, any tautology that appears in the list of axioms of an axiomatic system can be removed from the list without impairing the power of the axiomatic system.

An extreme case of an axiomatic system

An extreme case of an axiomatic system is _____. The axiomatic system for propositional logic is the _____. This means that all we can prove in propositional logic are _____.

Definition. An axiomatic system $a_1, a_2, \ldots, a_N$ is said to be *consistent* if there exists a proposition q which cannot be proven from the axioms. In symbols:

$$\exists\, q : \neg(a_1, a_2, \ldots, a_N \vdash q).$$

The idea behind this definition!

Consider an axiomatic system which contains contradicting propositions:

$$a_1, \ldots, s, \ldots, \neg s, \ldots, a_N.$$

Then, given any proposition q, the following is a proof of q within this system:

$$s, \neg s, q.$$

Indeed, s and $\neg s$ are legitimate steps in the proof since they are axioms. Moreover, $s \wedge \neg s$ is a contradiction and thus $(s \wedge \neg s) \Rightarrow q$ is a tautology. Therefore, q follows from condition (M). This shows that any proposition can be proven within a system with contradictory axioms. In other words, the inability to prove every proposition is a property possessed by no contradictory system, and hence we define a consistent system as one with this property.

Having come this far, we can now state (and prove) an impressively sounding theorem.

Theorem: *Propositional logic is consistent.*

Ex. *Write your proof*

From PSP, DSA, and ADA, you have already encountered many algorithms. But have you ever wondered what kinds of proofs we can actually write for them? One of the most fundamental proofs is showing that a program always produces the correct output—what we call **program correctness**.

You might say:

"I will run my program and check it!"

But this immediately raises the question:

"How many times?"

Is it possible to test every input? Clearly not! This is why it becomes extremely important to learn the **Art of Proofs**—a systematic and reliable way to demonstrate that your algorithm works for *all* valid inputs, not just the ones you test.

Most of your school-life experience with proofs probably came from **geometry**. So let's travel back to those familiar days and revisit two classic results.

Try to Prove:

Theorem: The sum of all interior angles of any triangle is 180°.

Ex. *Write your proof here.*

Theorem: An angle inscribed in a semicircle is always 90°.

Ex. *Write your proof here.*

These exercises help you reconnect with the foundations of geometric reasoning and prepare you for understanding more formal proofs in advanced mathematics.

5 Why Proofs Matter: Understanding Program Correctness

Consider the following program:

```
def odd_mult(n, m):
    # Check if both numbers are odd
    if n % 2 == 1 and m % 2 == 1:
        prod = n * m
    else:
        return False

    # Check if the product is odd
    if prod % 2 == 1:
        return True
    else:
        return False
```

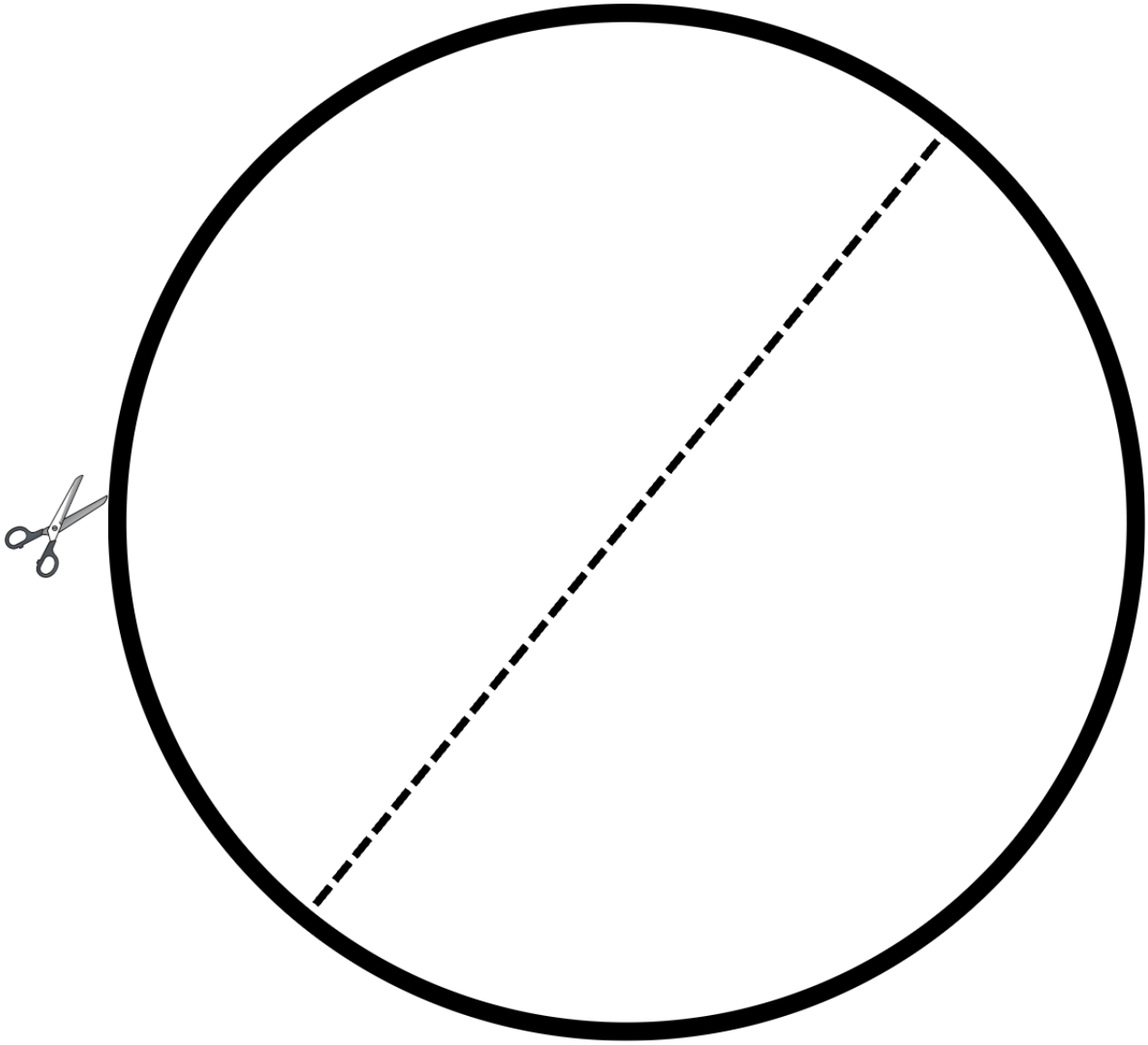
We want to show that whenever the user inputs two **odd** numbers, the program will **always** return `True`.

This is a useful example for demonstrating the importance of mathematical proof in computer science. Instead of presenting the abstract statement:

> **"Prove that the product of two odd numbers is odd,"**

we can **repackage** the same idea as a program correctness problem:

> "Prove that this program always returns `True` when both inputs are odd."

Ex. *Write your proof here.*

6 Activity